

# Finding Quadratic Equations and Using the Discriminant to Count X-Intercepts (Homework)

Use the discriminant to find the number of x-intercepts of the graph of each quadratic expression below. *Show all work or whoever reads your paper will assume that you copied the answers.*

①  $y = 5x^2 - 8x + 6$

⑤  $6x^2 - x + 2 = y$

②  $-x^2 + 2x - 10 = y$

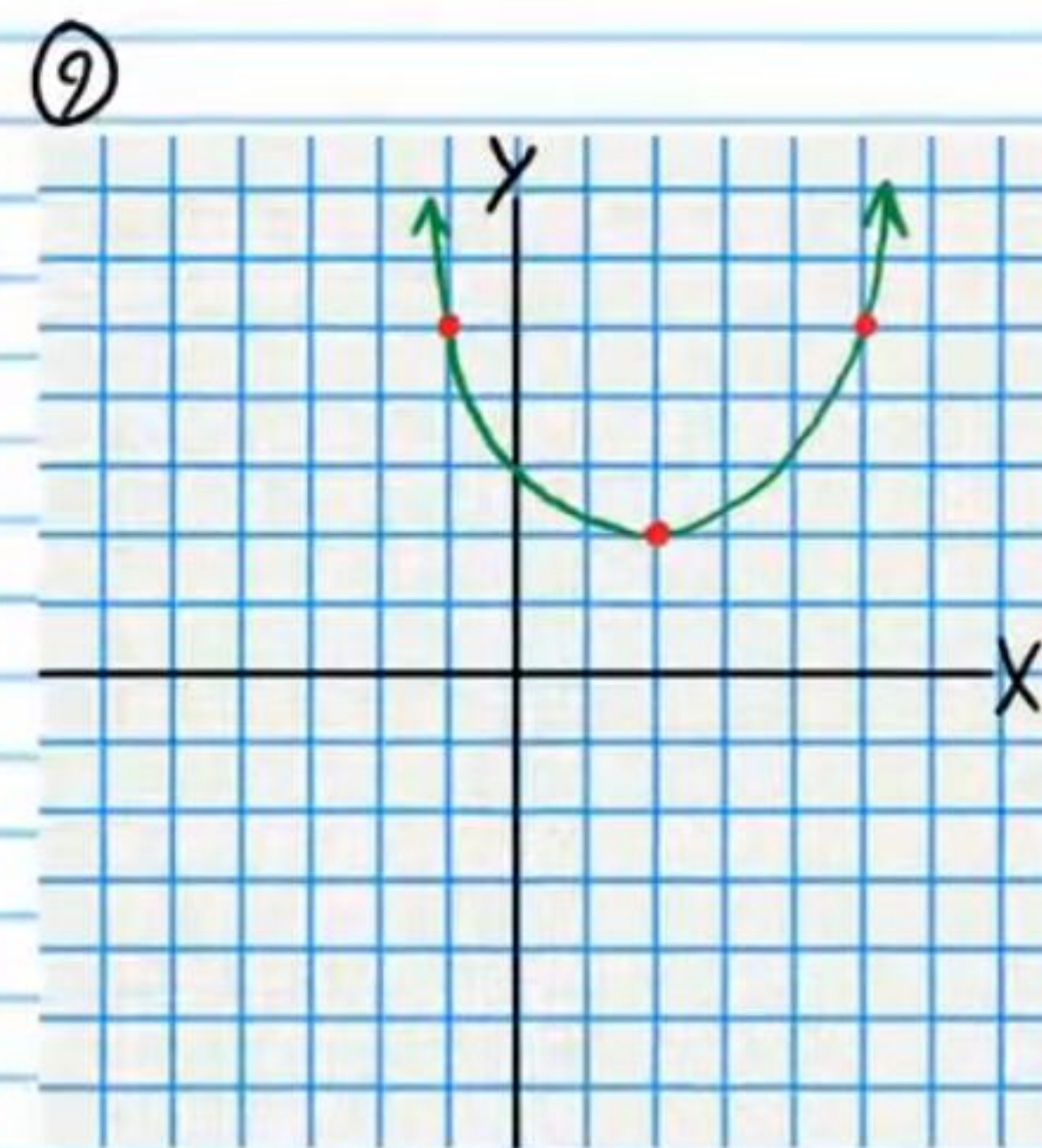
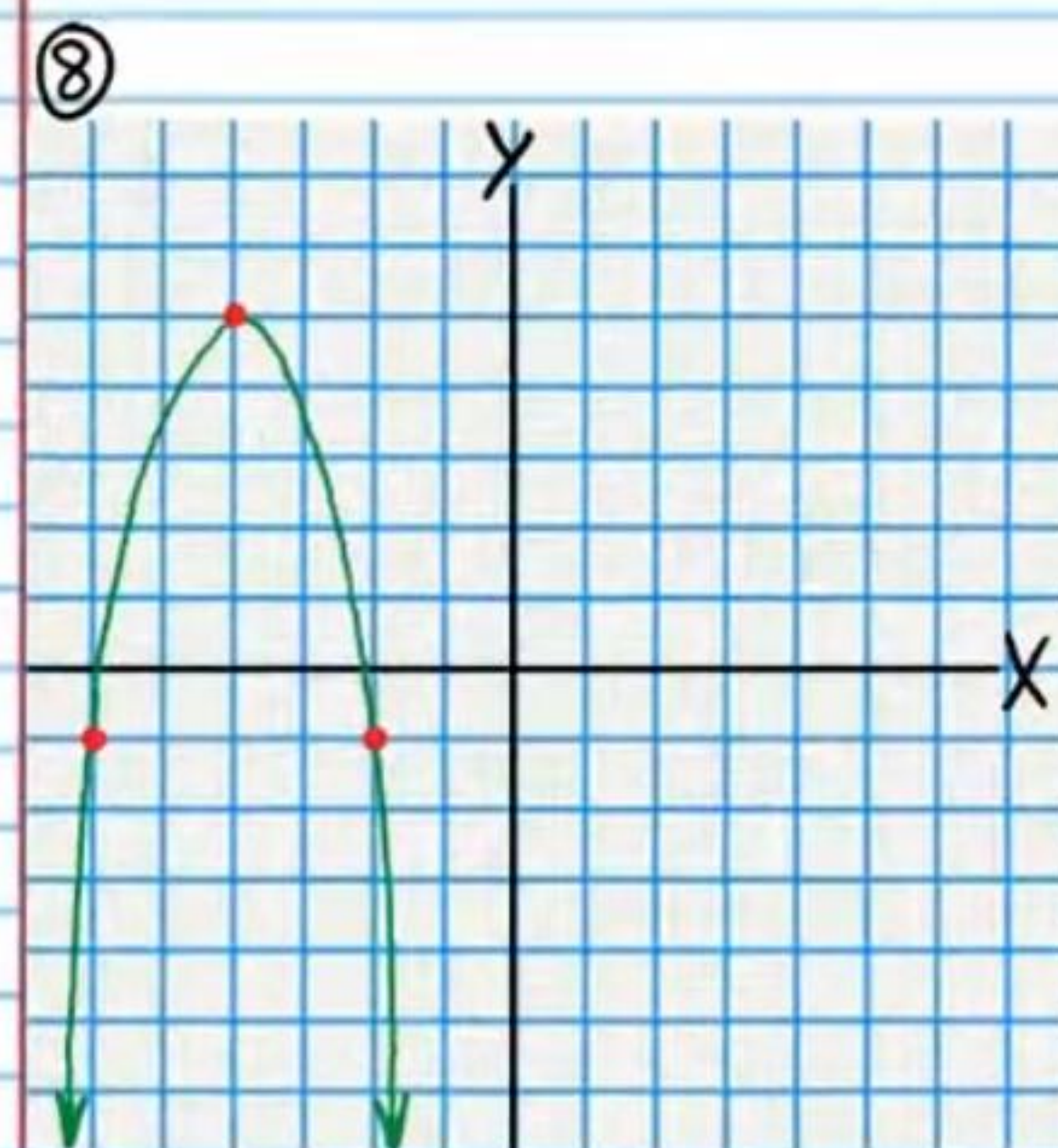
⑥  $9x^2 - 24x + 16 = y$

③  $y = x^2 + 2x + 1$

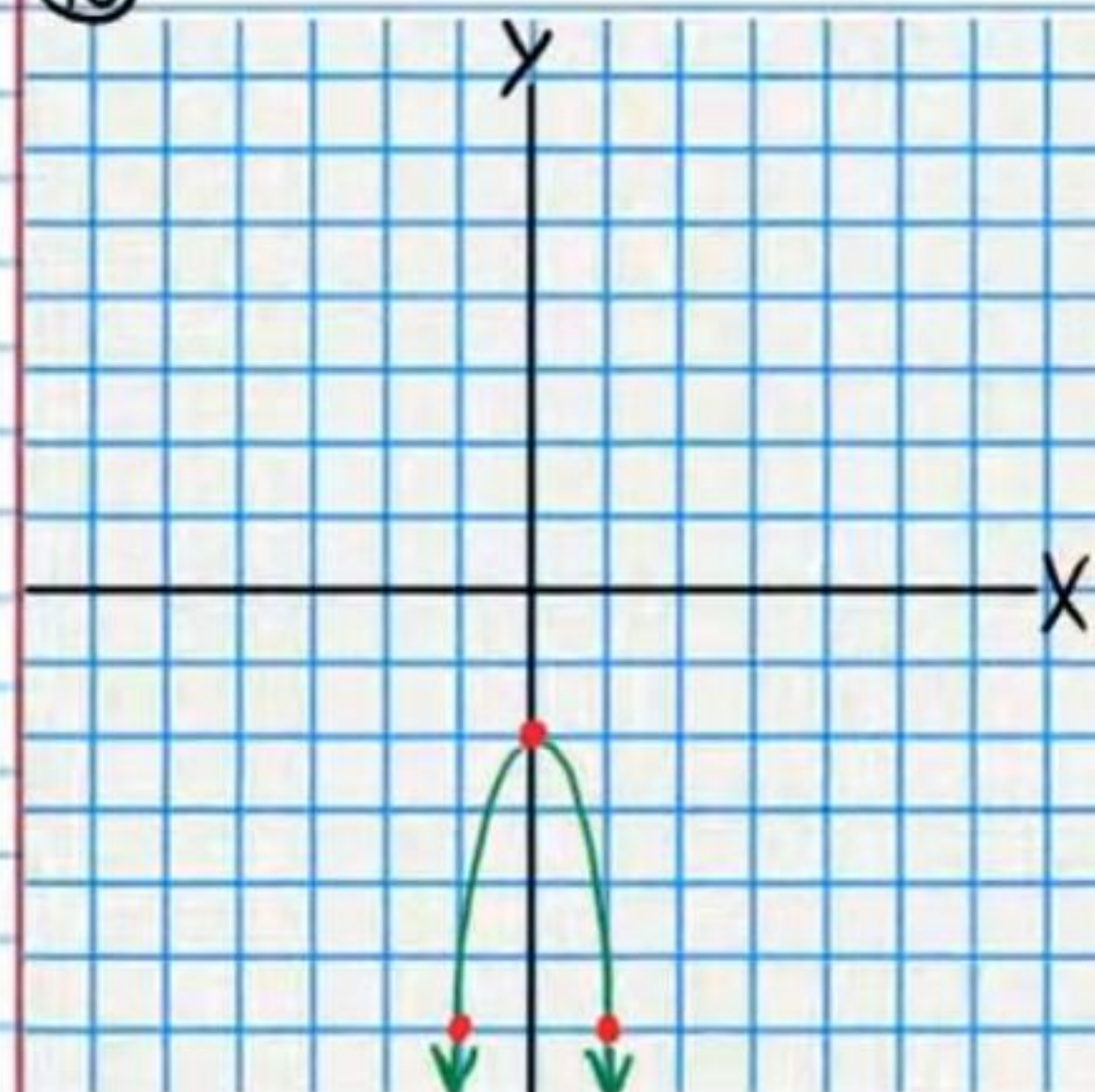
⑦  $y = -8x^2 - 3x + 5$

④  $y = -3x^2 + 4x + 7$

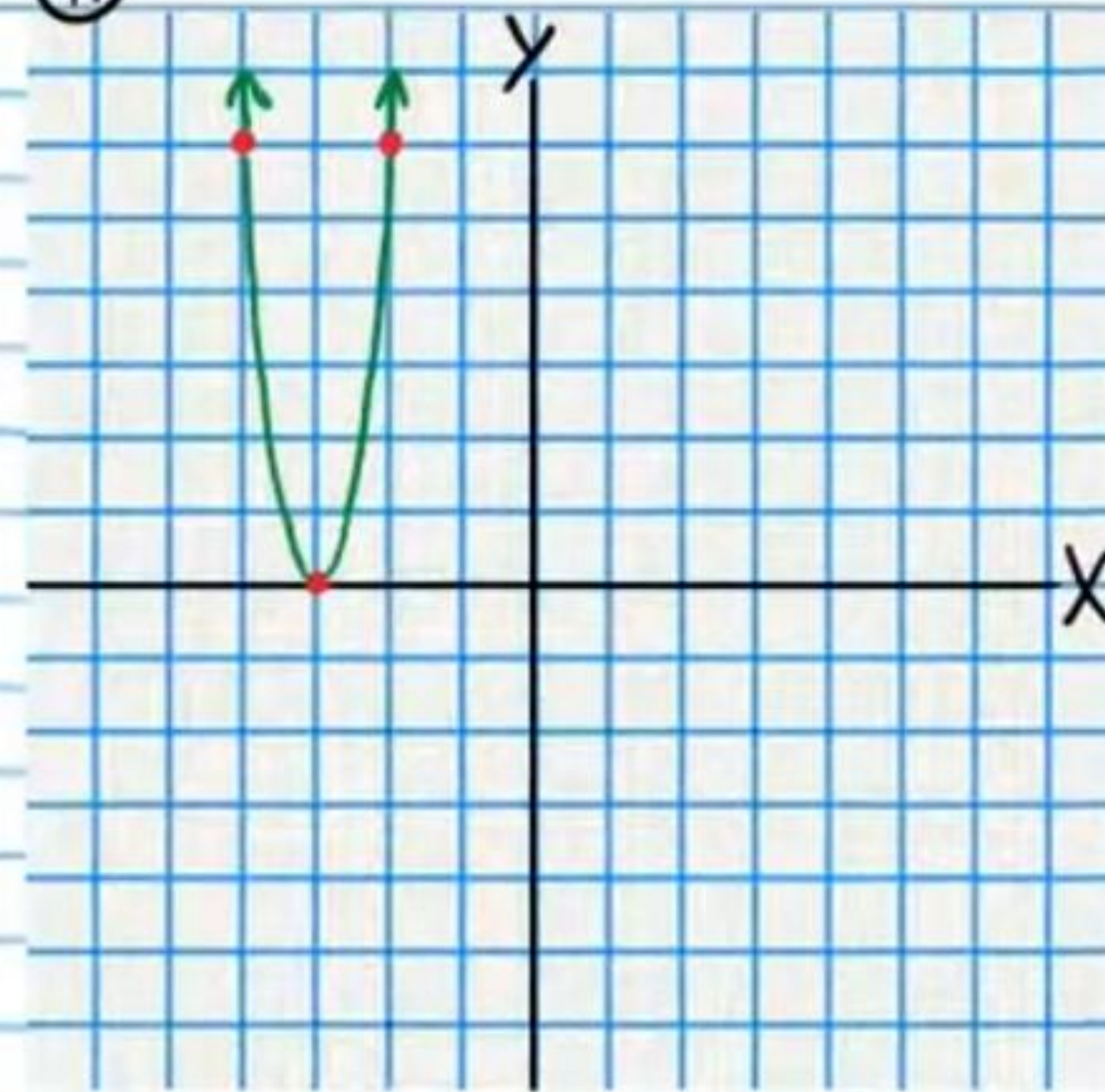
Write an equation of each parabola. *Hint: It is easiest to use vertex form.*



⑩



⑪



For each problem below, write an equation of the parabola with the given characteristics. *Hint: It is easiest to use vertex form. Assume that all corresponding parabolas open up or down.*

⑫ Vertex (5, 2) and point (-1, 4).

⑬ Vertex (-4, 0) and point (-2, -1).

⑭ Vertex (2, -3) and x-intercept 5.

⑮ Vertex (3, 8) and y-intercept -2.

⑯ Vertex (-4, -1) and directrix  $y = -5$ .

⑰ Vertex (-3, 2) and directrix  $y = 3$ .

⑱ Vertex (2, 1) and focus (2, 3).



- (19) Vertex  $(1, -4)$  and focus  $(1, -5)$ .  
 (20) Directrix  $y = 1$  and focus  $(2, 3)$ .  
 (21) Directrix  $y = -4$  and focus  $(-3, -7)$ .

## Answers

- (1) No x-intercepts  
 (2) No x-intercepts  
 (3) One x-intercept  
 (4) Two x-intercepts  
 (5) No x-intercepts  
 (6) One x-intercept  
 (7) Two x-intercepts  
 (8)  $y = -\frac{3}{2}(x+4)^2 + 5$   
 (9)  $y = \frac{1}{3}(x-2)^2 + 2$   
 (10)  $y = -4(x-0)^2 - 2$   
     simplify  $\Downarrow$   
      $y = -4x^2 - 2$   
 (11)  $y = 6(x+3)^2$   
 (12)  $y = \frac{1}{18}(x-5)^2 + 2$

- (13)  $y = -\frac{1}{4}(x+4)^2$   
 (14)  $y = \frac{1}{3}(x-2)^2 - 3$   
 (15)  $y = -\frac{10}{9}(x-3)^2 + 8$   
 (16)  $y = \frac{1}{16}(x+4)^2 - 1$   
 (17)  $y = -\frac{1}{4}(x+3)^2 + 2$   
 (18)  $y = \frac{1}{8}(x-2)^2 + 1$   
 (19)  $y = -\frac{1}{4}(x-1)^2 - 4$   
 (20)  $y = \frac{1}{4}(x-2)^2 + 2$   
 (21)  $y = -\frac{1}{6}(x+3)^2 - 5.5$